Synoptic Question 12: The Physics of Table Tennis

Reading Passage

Read the following article and then answer the questions that follow.

Table tennis is a sport characterised by extremely high speeds and reaction times. A modern competitive player can hit the ball at speeds over 110 km/h, covering the length of the table in a fraction of a second. The game is played on a 2.74 m long table divided by a 15.25 cm high net.

The interaction between the paddle (or "bat") and the ball is a complex collision. The paddles are covered with a layer of rubber, which allows players to impart massive amounts of spin to the ball. This spin is critical to the game. A ball with topspin experiences a downward-curving trajectory due to an aerodynamic force known as the Magnus effect. This allows players to hit the ball very fast while still having it "dip" and land on the table. Conversely, backspin causes the ball to "float" and take a flatter trajectory.

The ball itself is very light and has a large surface area for its mass, making it highly susceptible to air resistance (drag). This drag force is a significant factor in the ball's deceleration during flight. When the ball bounces, it loses kinetic energy, and the spin can "bite" the table, causing the ball to change its horizontal speed and direction due to friction.

Data:

Data for a typical table tennis setup and play:

Table 1: Table Tennis Data

Parameter	**Value**
Mass of table tennis ball, m	2.70 g
Diameter of ball, d	40.0 mm
Length of table, L	2.74 m
Height of net, h_{net}	0.1525 m
Speed of a "smash", v_s	110 km/h
Initial vertical position of smash	0.30 m above the table
Horizontal distance from player for smash	0.50 m from end of table
Contact time between paddle and ball, Δt	3.0 ms
Density of air, ρ	1.22 kg m^{-3}
Drag coefficient of a sphere, C_D	0.50
Coefficient of kinetic friction (table-ball), μ_k	0.25

Questions

(a) A player hits a smash. The ball is stationary at the peak of its toss (0.30 m) and is struck by the paddle.

  (i) Convert the ball's mass to kg and its smash speed (110 km/h) to m s^{-1}.

  (ii) Calculate the impulse delivered to the ball to achieve this speed.

  (iii) Calculate the average force exerted by the paddle on the ball during the 3.0 ms contact time.

(b) Consider the energy transfer during the smash.

  (i) Calculate the kinetic energy of the ball just after impact.

  (ii) Calculate the average power delivered to the ball by the paddle.

(c) Let's analyse the trajectory of the smash. Assume the ball is hit perfectly horizontally from a height of 0.30 m and a horizontal distance of 0.50 m from the player's end of the table. **For this part (c) only, ignore air resistance and spin.**

  (i) The net is 1.37 m from the player's end of the table. Calculate the time it takes the ball to reach the net.

  (ii) Calculate the vertical distance the ball drops due to gravity by the time it reaches the net.

(iii) The net is 0.1525 m high. Determine if the smash successfully clears the net.

(d) The effects of air resistance (drag) are significant. The drag force F_D is given by $F_D = \frac{1}{2}\rho C_D A v^2$, where A is the cross-sectional area.

(i) Calculate the cross-sectional area of the ball.

(ii) Calculate the initial drag force on the ball at its smash speed.

(iii) Calculate the weight of the ball and compare it to the initial drag force.

(e) A player imparts topspin to the ball.

(i) Draw a simple diagram of a spinning ball moving through the air. Show the direction of rotation (topspin), the direction of air flow relative to the top and bottom surfaces, and the direction of the resultant Magnus force.

(ii) Explain, with reference to air pressure or velocity, why this spin creates a downward force (the Magnus effect).

(f) When a spinning ball bounces, friction plays a key role. A ball with heavy topspin is moving horizontally to the right and strikes the table.

(i) What is the direction of the frictional force exerted **by the table on the ball**?

(ii) How does this frictional force affect the horizontal velocity of the ball after the bounce?

Solutions

(a) (i) Mass $m = 2.70$ g $= 0.00270$ kg.
Speed $v_s = 110 \text{ km/h} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 30.56 \text{ m s}^{-1}$.

(ii) Impulse $= \Delta p = m(v_f - v_i) = (0.00270)(30.56 - 0) = 0.0825$ Ns.

(iii) $\Delta t = 3.0$ ms $= 0.0030$ s.
$F_{avg} = \frac{\Delta p}{\Delta t} = \frac{0.0825}{0.0030} = 27.5$ N.

(b) (i) $KE = \frac{1}{2}mv_s^2 = \frac{1}{2}(0.00270)(30.56)^2 = 1.26$ J.

(ii) $P_{avg} = \frac{\text{Work Done}}{\text{Time}} = \frac{\Delta KE}{\Delta t} = \frac{1.26}{0.0030} = 420$ W.

(c) (i) Horizontal distance to net $= 1.37$ m $- 0.50$ m $= 0.87$ m.
Time to net $t_{net} = \frac{\text{distance}}{v_s} = \frac{0.87}{30.56} = 0.0285$ s.

(ii) Vertical drop $s_y = u_y t + \frac{1}{2}a_y t^2 = 0 + \frac{1}{2}(9.81)(0.0285)^2 = 0.00398$ m.

(iii) Height at net = Initial height - drop $= 0.30$ m $- 0.00398$ m $= 0.296$ m.
Since $0.296 \text{ m} > 0.1525$ m (net height), the ball easily clears the net.

(d) (i) Radius $r = d/2 = 20.0$ mm $= 0.020$ m.
Area $A = \pi r^2 = \pi(0.020)^2 = 0.001257 \text{ m}^2$.

(ii) $F_D = \frac{1}{2}\rho C_D A v^2 = \frac{1}{2}(1.22)(0.50)(0.001257)(30.56)^2 = 0.358$ N.

(iii) Weight $W = mg = (0.00270)(9.81) = 0.0265$ N.
The initial drag force (0.358 N) is about 13.5 times larger than the ball's weight (0.0265 N). This shows air resistance is the dominant force.

(e) (i)

Diagram of Magnus Effect (Topspin)

- Ball moves left to right.
- Ball rotates clockwise (topspin).
- **Top surface:** Air flow is faster (rotation is against airflow).
- **Bottom surface:** Air flow is slower (rotation is with airflow).
- **Resultant Force (Magnus):** Downwards.

(ii) Due to topspin, the air speed relative to the top surface of the ball is higher than the air speed relative to the bottom surface. According to Bernoulli's principle, higher-speed air has lower pressure. This creates a pressure difference, with lower pressure on top and higher pressure on the bottom, resulting in a net downward force (the Magnus force) that causes the ball to dip.

(f) (i) With topspin, the bottom surface of the ball is moving to the right (in the direction of travel) relative to the ball's center. When it hits the table, this surface tries to drag the table to the right. By Newton's third law, the table exerts a frictional force on the ball **to the left** (opposite to the motion of the contact point).

(ii) This leftward (backward) frictional force opposes the ball's forward horizontal velocity. This causes the ball to slow down horizontally more than it would without spin and to "kick" upwards with a higher vertical bounce.

Variant 2 (Higher Difficulty) — CIE P4 Style

Data for this question

Parameter	Value
Mass of ball, m	$2.70 \times 10^{-3}\,\mathrm{kg}$
Diameter of ball, d	$4.00 \times 10^{-2}\,\mathrm{m}$
Cross-sectional area of ball, A	$1.26 \times 10^{-3}\,\mathrm{m}^2$
Table length	$2.74\,\mathrm{m}$
Net height	$0.1525\,\mathrm{m}$
Air density, ρ	$1.22\,\mathrm{kg\,m^{-3}}$
Drag coefficient, C_D	0.50
Gravitational field strength, g	$9.81\,\mathrm{m\,s^{-2}}$
Coefficient of restitution (ball/table), e	0.82

A player performs an aggressive topspin forehand from 0.45 m above the table at a point 0.30 m behind the end line. At impact, the ball speed is $33.0\,\mathrm{m\,s^{-1}}$ at 8.0° below the horizontal.

Question 1

(a) Resolve the initial velocity into horizontal and vertical components. Determine the time taken for the ball to reach the net plane (1.37 m from the player's end line), neglecting drag for this part only. [3]

(b) Show whether the ball clears the net when drag is neglected. [3]

(c) The horizontal drag force is modelled by

$$F_D = \frac{1}{2}\rho C_D A v_x^2.$$

Using the horizontal speed from part (a), calculate the initial drag force and the corresponding initial horizontal deceleration. Comment on the significance of this deceleration compared with g. [4]

(d) **Open-ended concept question.** Explain why topspin allows a player to strike the ball harder while still keeping it on the table. Your answer should include: relative airflow, pressure difference, force direction, and trajectory shape. [5]

(e) The ball reaches the opponent with horizontal speed $20.0\,\mathrm{m\,s^{-1}}$ and vertical speed $-3.6\,\mathrm{m\,s^{-1}}$ just before bounce.

(i) Determine the vertical speed immediately after bounce. [2]

(ii) If kinetic friction at the table acts for 2.5 ms with coefficient $\mu = 0.28$, estimate the change in horizontal speed during contact. [3]

(iii) Suggest one reason why the true change in horizontal speed may differ from your estimate. [1]

(f) **Open-ended evaluation question.** A coach claims: "For elite table tennis, drag is more important than weight in determining flight." Evaluate this statement quantitatively using appropriate values, and discuss one limitation of this comparison. [5]

Total: 26 marks

Sample Solution (Variant 2)

(a)

$$v_x = 33.0\cos 8.0^\circ = 32.7\,\mathrm{m\,s^{-1}}, \qquad v_y = -33.0\sin 8.0^\circ = -4.59\,\mathrm{m\,s^{-1}}.$$

Horizontal distance from impact point to net plane:

$$x = 0.30 + 1.37 = 1.67\,\mathrm{m}.$$

Time to net:

$$t = \frac{x}{v_x} = \frac{1.67}{32.7} = 5.11 \times 10^{-2}\,\mathrm{s}.$$

(b) Vertical displacement at net:

$$y = u_y t + \frac{1}{2}(-g)t^2 = (-4.59)(0.0511) - \frac{1}{2}(9.81)(0.0511)^2 = -0.247\,\mathrm{m}.$$

Height above table at net:

$$h = 0.45 + y = 0.45 - 0.247 = 0.203\,\mathrm{m}.$$

Since $0.203\,\mathrm{m} > 0.1525\,\mathrm{m}$, the ball clears the net by about $0.050\,\mathrm{m}$.

(c)

$$F_D = \frac{1}{2}(1.22)(0.50)(1.26 \times 10^{-3})(32.7)^2 = 0.412\,\mathrm{N}.$$

$$a_x = \frac{F_D}{m} = \frac{0.412}{2.70 \times 10^{-3}} \approx 1.53 \times 10^2\,\mathrm{m\,s^{-2}}.$$

So $a_x \approx 15.6g$, indicating very strong horizontal slowing.

(d) Topspin means the top of the ball moves against the local airflow while the bottom moves with it. Relative speed is larger over one side and smaller over the other, producing a pressure difference. The net aerodynamic lift (Magnus force) acts downward for topspin. This downward force adds to the vertical acceleration, so the trajectory dips more quickly than under gravity alone. Therefore a player can hit with greater forward speed yet still make the ball land before the end line.

(e) (i) With $e = \dfrac{v_{y,\text{after}}}{|v_{y,\text{before}}|}$,

$$v_{y,\text{after}} = e|v_{y,\text{before}}| = 0.82 \times 3.6 = 2.95\,\mathrm{m\,s^{-1}} \text{ upward}.$$

(ii) Normal reaction during contact is approximately $N \approx mg$, giving friction

$$F_f \approx \mu mg = 0.28(2.70 \times 10^{-3})(9.81) = 7.42 \times 10^{-3}\,\mathrm{N}.$$

$$a_f = \frac{F_f}{m} = 2.75\,\mathrm{m\,s^{-2}}, \quad \Delta v_x = a_f \Delta t = 2.75(2.5 \times 10^{-3}) = 6.9 \times 10^{-3}\,\mathrm{m\,s^{-1}}.$$

Speed decreases by about $0.007\,\mathrm{m\,s^{-1}}$ in this crude model.

(iii) Real contact force and friction vary rapidly during deformation, and spin changes slipping/rolling conditions; hence constant-friction modelling is limited.

(f) Weight:

$$W = mg = (2.70 \times 10^{-3})(9.81) = 0.0265\,\mathrm{N}.$$

Using part (c), initial drag is about 0.412 N. Ratio:

$$\frac{F_D}{W} \approx \frac{0.412}{0.0265} \approx 15.5.$$

Quantitatively, drag is much larger than weight at high speed, so the coach's claim is valid for early flight after a powerful stroke. Limitation: drag decreases strongly as speed falls ($\propto v^2$), so its dominance changes during the trajectory; spin-dependent Magnus effects are also excluded in a simple drag-vs-weight comparison.

Mark Scheme (CIE P4 Style, Variant 2)

General marking principles (brief)

- Award marks for correct physics, clear method, and valid units/significant figures.
- Ecf (error carried forward) may be applied where appropriate.
- For open-ended parts, credit coherent physics arguments using accepted terminology.

Detailed mark allocation

3
- Correct component method using trigonometric resolution (1)
- $v_x \approx 32.7\,\mathrm{m\,s^{-1}}$ and $v_y \approx -4.59\,\mathrm{m\,s^{-1}}$ (1)
- Correct time to net using $t = x/v_x$, with $x = 1.67\,\mathrm{m}$, giving $t \approx 0.051\,\mathrm{s}$ (1)

3
- Correct vertical-motion expression with sign convention (1)
- Height at net correctly obtained (about $0.203\,\mathrm{m}$) (1)
- Clear comparison with net height and correct conclusion (clears net) (1)

4
- Correct substitution into drag equation (1)
- $F_D \approx 0.41\,\mathrm{N}$ (allow range due to rounding) (1)
- Correct deceleration calculation $a_x = F_D/m \approx 1.5 \times 10^2\,\mathrm{m\,s^{-2}}$ (1)
- Valid comparison with g (about $15g$) and significance comment (1)

5 (open-ended)

- Describes differential relative airflow around spinning ball (1)
- Links airflow difference to pressure difference / circulation argument (1)
- Identifies Magnus force direction as downward for topspin (1)
- Explains steeper downward curvature / faster dip of trajectory (1)
- Concludes this permits higher striking speed while landing on table (1)

(a) (i) [2]

- Uses $e = v_{after}/v_{before}$ relation correctly in normal direction (1)
- $v_{y,after} \approx 2.95\,\mathrm{m\,s^{-1}}$ upward (1)

(b) (ii) [3]

- Uses friction model $F = \mu N$ with suitable assumption for N (1)
- Obtains acceleration then $\Delta v_x = a\Delta t$ (1)
- Numerical estimate with correct direction (speed reduction) (1)

(c) (iii) [1]

- One valid modelling limitation, e.g. variable contact force, spin/slip transition, changing μ (1)

5 (open-ended evaluation)

- Calculates/quotes weight correctly (1)
- Calculates/quotes drag magnitude correctly (1)
- Compares with sensible ratio or order-of-magnitude statement (1)
- Makes justified judgement about claim validity in context (1)
- Gives one clear limitation/qualification (speed dependence, spin effects, trajectory stage) (1)